CP3 - DBS - Vytvoření databáze

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DDL SQL pro vytvoření databáze

```sql
DROP TABLE IF EXISTS "UcastniSe" CASCADE;
DROP TABLE IF EXISTS "ZaznamDochazky" CASCADE;
DROP TABLE IF EXISTS "Externista" CASCADE;
DROP TABLE IF EXISTS "Manazer" CASCADE;
DROP TABLE IF EXISTS "Adresa" CASCADE;
DROP TABLE IF EXISTS "Telefon" CASCADE;
DROP TABLE IF EXISTS "Zamestnanec" CASCADE;
DROP TABLE IF EXISTS "Projekt" CASCADE;
DROP TABLE IF EXISTS "Pozice" CASCADE;
DROP TABLE IF EXISTS "Oddeleni" CASCADE;


CREATE TABLE "Oddeleni" (
  "id_oddeleni" int4 NOT NULL,
  "cislo_oddeleni" varchar(6) NOT NULL,
  "nazev" varchar(100) NOT NULL,
  "lokace" varchar(100) NOT NULL,
  "popis" text,
  "id_nadrizene_oddeleni" int4,
  PRIMARY KEY("id_oddeleni"),
  CONSTRAINT "unique_cislo_oddeleni" UNIQUE("cislo_oddeleni"),
  CONSTRAINT "unique_nazev_lokace" UNIQUE("nazev","lokace"),
  CONSTRAINT "Ref_Oddeleni_to_Oddeleni" FOREIGN KEY ("id_nadrizene_oddeleni")
    REFERENCES "Oddeleni"("id_oddeleni")
    MATCH SIMPLE
    ON DELETE SET NULL
    ON UPDATE NO ACTION
    NOT DEFERRABLE
);

CREATE TABLE "Pozice" (
  "id_pozice" int4 NOT NULL,
  "nazev" varchar(100) NOT NULL,
  "uroven" int4 NOT NULL,
  "popis" text,
  PRIMARY KEY("id_pozice"),
  CONSTRAINT "unique_nazev" UNIQUE("nazev"),
  CONSTRAINT "check_validni_uroven" CHECK(uroven BETWEEN 1 AND 10)
);

CREATE TABLE "Projekt" (
  "id_projekt" int4 NOT NULL,
  "nazev" varchar(64) NOT NULL,
  "termin_zahajeni" date NOT NULL,
  "termin_ukonceni" date,
  "popis" text,
  PRIMARY KEY("id_projekt"),
  CONSTRAINT "unique_nazev_termin_zahajeni" UNIQUE("nazev","termin_zahajeni"),
  CONSTRAINT "check_termin_zahajeni_ukonceni" CHECK(termin_ukonceni >=
termin_zahajeni)
);
```

```sql
CREATE TABLE "Zamestnanec" (
  "id_zamestnanec" int4 NOT NULL,
  "rodne_cislo" varchar(11) NOT NULL,
  "email" varchar(100) NOT NULL,
  "jmeno" varchar(100) NOT NULL,
  "prijmeni" varchar(100) NOT NULL,
  "datum_nastupu" date NOT NULL,
  "id_oddeleni" int4 NOT NULL,
  "id_pozice" int4,
  PRIMARY KEY("id_zamestnanec"),
  CONSTRAINT "unique_rodne_cislo" UNIQUE("rodne_cislo"),
  CONSTRAINT "unique_email" UNIQUE("email"),
  CONSTRAINT "unique_zamestnanec_oddeleni"
UNIQUE("id_zamestnanec","id_oddeleni"),
  CONSTRAINT "check_email_format" CHECK(email ~* '^[A-Za-z0-9._%+-]+@[A-Za-
z0-9.-]+\.[A-Za-z]{2,}$'),
  CONSTRAINT "check_rodne_cislo_format" CHECK(rodne_cislo ~ '^[0-9]{6}/[0-9]
{3,4}$'),
  CONSTRAINT "Ref_Zamestnanec_to_Oddeleni" FOREIGN KEY ("id_oddeleni")
    REFERENCES "Oddeleni"("id_oddeleni")
    MATCH SIMPLE
    ON DELETE RESTRICT
    ON UPDATE CASCADE
    NOT DEFERRABLE,
  CONSTRAINT "Ref_Zamestnanec_to_Pozice" FOREIGN KEY ("id_pozice")
    REFERENCES "Pozice"("id_pozice")
    MATCH SIMPLE
    ON DELETE SET NULL
    ON UPDATE CASCADE
    NOT DEFERRABLE
);

CREATE TABLE "Telefon" (
  "id_telefon" int4 NOT NULL,
  "telefon" varchar NOT NULL,
  "id_zamestnanec" int4 NOT NULL,
  PRIMARY KEY("id_telefon"),
  CONSTRAINT "unique_telefon_zamestnanec" UNIQUE("telefon","id_zamestnanec"),
  CONSTRAINT "check_validni_telefon" CHECK(telefon ~ '^\+?[0-9\s]{9,15}$'),
  CONSTRAINT "Ref_Telefon_to_Zamestnanec" FOREIGN KEY ("id_zamestnanec")
    REFERENCES "Zamestnanec"("id_zamestnanec")
    MATCH SIMPLE
    ON DELETE CASCADE
    ON UPDATE CASCADE
    NOT DEFERRABLE
);

CREATE TABLE "Adresa" (
  "id_adresa" int4 NOT NULL,
  "ulice" varchar NOT NULL,
  "mesto" varchar NOT NULL,
  "psc" varchar NOT NULL,
```

```
  "id_zamestnanec" int4 NOT NULL,
  PRIMARY KEY("id_adresa"),
  CONSTRAINT "unique_ulice_mesto_psc_zamestnanec"
UNIQUE("ulice","mesto","psc","id_zamestnanec"),
  CONSTRAINT "check_validni_psc" CHECK(psc ~ '^[0-9]{3}\s?[0-9]{2}$'),
  CONSTRAINT "Ref_Adresa_to_Zamestnanec" FOREIGN KEY ("id_zamestnanec")
    REFERENCES "Zamestnanec"("id_zamestnanec")
    MATCH SIMPLE
    ON DELETE CASCADE
    ON UPDATE CASCADE
    NOT DEFERRABLE
);

CREATE TABLE "Manazer" (
  "id_manazer" int4 NOT NULL,
  "uroven_pravomoci" int4 NOT NULL,
  "id_zamestnanec" int4 NOT NULL,
  "id_oddeleni" int4 NOT NULL,
  PRIMARY KEY("id_manazer"),
  CONSTRAINT "unique_manazer_zamestnanec" UNIQUE("id_zamestnanec"),
  CONSTRAINT "check_validni_uroven_pravomoci" CHECK(uroven_pravomoci BETWEEN 1
AND 5),
  CONSTRAINT "Ref_Manazer_to_Zamestnanec" FOREIGN KEY ("id_zamestnanec",
"id_oddeleni")
    REFERENCES "Zamestnanec"("id_zamestnanec", "id_oddeleni")
    MATCH SIMPLE
    ON DELETE CASCADE
    ON UPDATE CASCADE
    NOT DEFERRABLE
);

CREATE TABLE "Externista" (
  "id_externista" int4 NOT NULL,
  "nazev_agentury" varchar NOT NULL,
  "konec_smlouvy" date NOT NULL,
  "id_zamestnanec" int4 NOT NULL,
  PRIMARY KEY("id_externista"),
  CONSTRAINT "unique_externista_zamestnanec" UNIQUE("id_zamestnanec"),
  CONSTRAINT "Ref_Externista_to_Zamestnanec" FOREIGN KEY ("id_zamestnanec")
    REFERENCES "Zamestnanec"("id_zamestnanec")
    MATCH SIMPLE
    ON DELETE CASCADE
    ON UPDATE CASCADE
    NOT DEFERRABLE
);

CREATE TABLE "ZaznamDochazky" (
  "id_dochazka" int4 NOT NULL,
  "cas_prichodu" timestamp NOT NULL,
  "cas_odchodu" timestamp,
  "typ_zaznamu" varchar(100) NOT NULL,
  "id_zamestnanec" int4 NOT NULL,
  PRIMARY KEY("id_dochazka"),
```

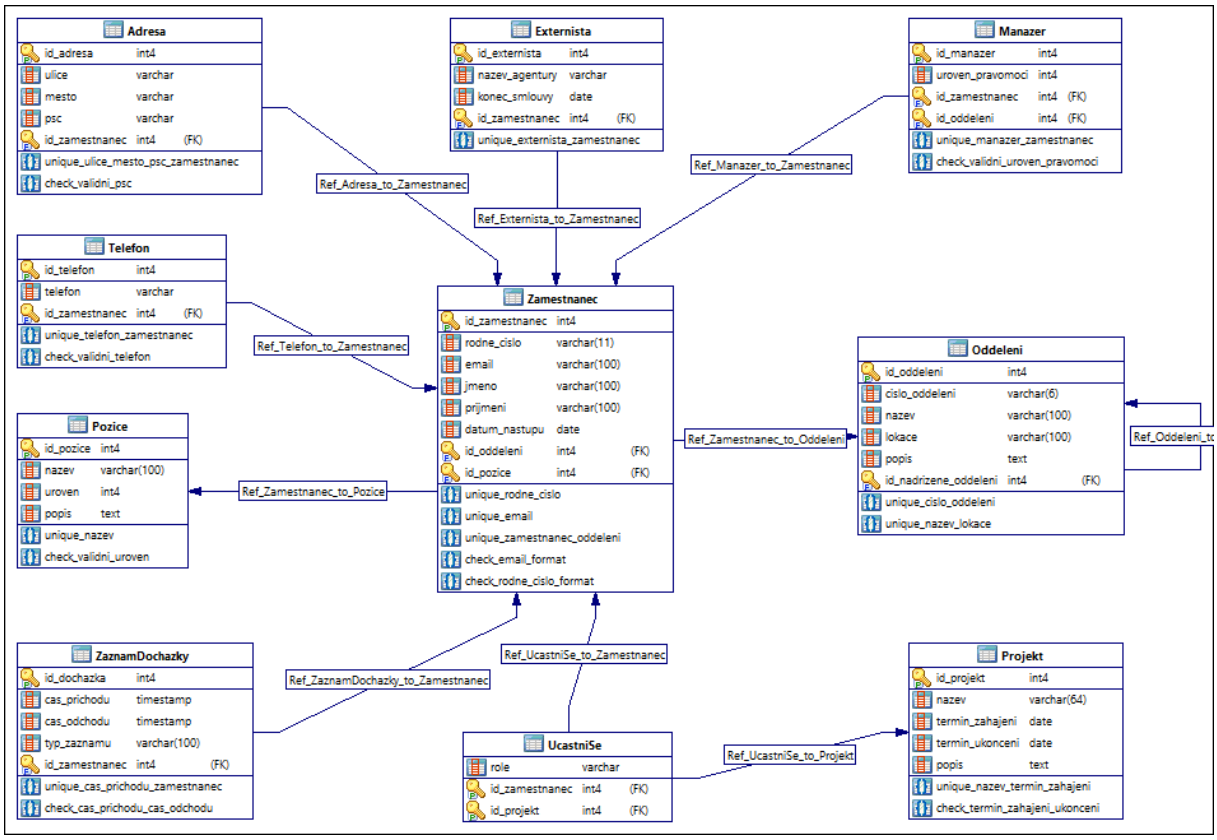
```sql
  CONSTRAINT "unique_cas_prichodu_zamestnanec"
UNIQUE("cas_prichodu","id_zamestnanec"),
  CONSTRAINT "check_cas_prichodu_cas_odchodu" CHECK(cas_odchodu > cas_prichodu),
  CONSTRAINT "Ref_ZaznamDochazky_to_Zamestnanec" FOREIGN KEY ("id_zamestnanec")
    REFERENCES "Zamestnanec"("id_zamestnanec")
    MATCH SIMPLE
    ON DELETE CASCADE
    ON UPDATE CASCADE
    NOT DEFERRABLE
);

CREATE TABLE "UcastniSe" (
  "role" varchar NOT NULL,
  "id_zamestnanec" int4 NOT NULL,
  "id_projekt" int4 NOT NULL,
  PRIMARY KEY("id_zamestnanec","id_projekt"),
  CONSTRAINT "Ref_UcastniSe_to_Zamestnanec" FOREIGN KEY ("id_zamestnanec")
    REFERENCES "Zamestnanec"("id_zamestnanec")
    MATCH SIMPLE
    ON DELETE CASCADE
    ON UPDATE CASCADE
    NOT DEFERRABLE,
  CONSTRAINT "Ref_UcastniSe_to_Projekt" FOREIGN KEY ("id_projekt")
    REFERENCES "Projekt"("id_projekt")
    MATCH SIMPLE
    ON DELETE CASCADE
    ON UPDATE CASCADE
    NOT DEFERRABLE
);
```

SQL 1: SQL DDL příkazy pro vytvoření databáze

Relační diagram

ER diagram



Obrázek 1: ER diagram

SQL query dotazy